Predictive Modelling of Washington D.C. Daily Bike Rentals

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Introduction

In this report, we will detail the development of a predictive linear model for the response variable, total daily count of rented bikes (cnt) located in Washington D.C.. In our analysis, we will draw upon data from the Capital Bike share System between 2011 and 2012 given in the coursework brief. Exploring the demand for bike sharing systems is crucial as it plays a significant role in traffic congestion, addressing environmental and health problems.

We know that regression analysis aims to exploit the pattern of correlations in the data to provide simple models that have explanatory and/or predictive power (Section 1.1). To be able to use this, in this report, we will restrict our analysis to the class of normal linear models (Section 1.3) in which the structural relationship is a linear combination of the unknown parameters.

This analysis relies on a data set of 550 daily observations. This data set consists of our response variable (cnt) and a vector of covariates for each case, including true continuous measurements such as temperature, wind speed and humidity. It also has categorical variables like the day of the week, season and weather situation. We will treat these categorical covariates as factors in R so that it can appropriately code their levels using dummy variables (Section 1.6).

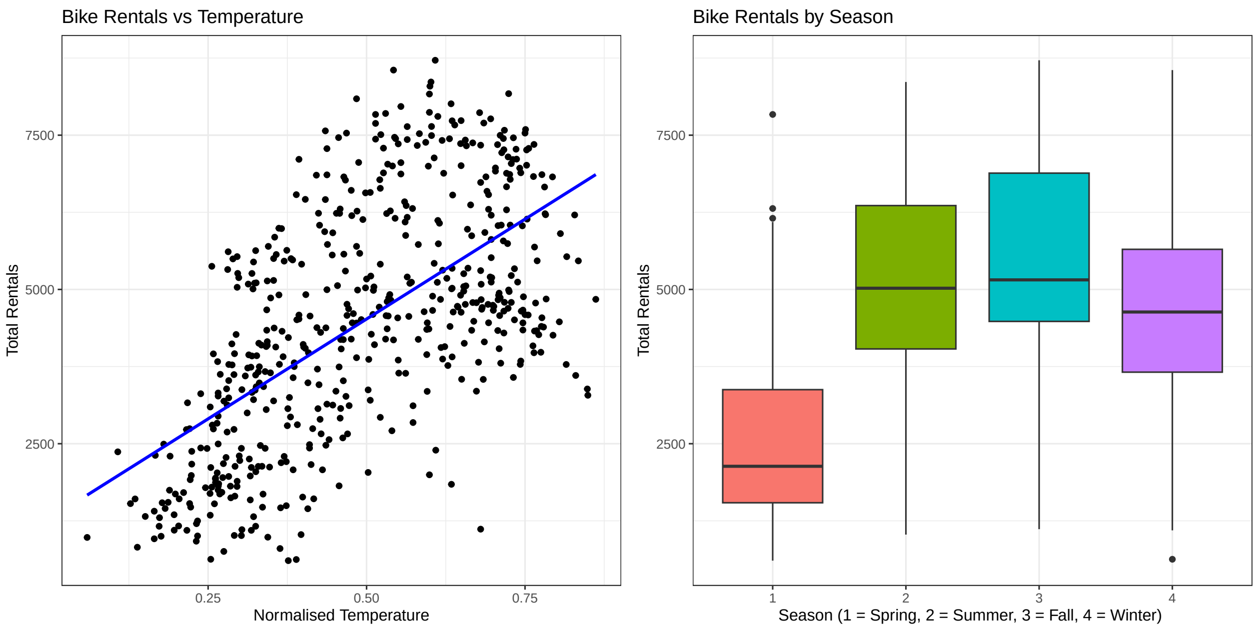
This report will be structured into two primary sections:

- Part(a)- We begin this section by conducting an exploratory data analysis (EDA) so that we can suggest sensible models with the use of box-plots and scatter plots. Following this, we will perform a rigorous model selection in order to find the best parsimonious multiple linear regression model. Finally, we will check our model assumptions using plots such as Residual vs Fitted, Q-Q residual, Scale-Location and Fitted values plots. We will then be able to interpret the estimated parameters to understand which variables most strongly affect the demand for daily bike rentals.
- Part(b)- In this section, we will test our parsimonious model with the new, unseen data (BikeTest.csv). We will calculate Test Deviance scores and generate scatter graphs to evaluate our models against a simple baseline model (cnt ~ season).

Part (a):

Exploratory Analysis

As seen in the introduction stage, we will begin by conducting an exploratory data analysis. From this we will gain a general picture of the data, identifying the strongest predictors of bike demands and spot any potential non-linear relationships prior to the formal model fitting.



Discussion on EDA: The diagrams above provide a clear picture of how temporal and environmental factors influence the daily count of rental bikes (cnt).

- Temperatures: The scatter graph above demonstrates a fairly strong positive correlation between normalised temperature and the daily count of rental bikes. We can see that as the temperature rises the amount of demand for rental bikes increases constantly. A transformation is not necessary as there does not appear to be a severe non-linear trend to indicate one.
- Seasons: The above boxplots reveal there is significant variation in demand for rental bikes in comparison between different seasons. We can see that the median number of bikes rented out during spring is by far the lowest, with the median peaking during the fall and summer months.

These findings indicate that both temporal and seasonal factors are highly associated with the daily count of rental bikes (the response variable). This justifies their inclusion as covariates in the proposed multiple linear regression model.

Model Selection

Since we have established the general picture of the data in the diagrams from the EDA, we now seek to develop a predictive linear regression model. The goal is to find the most parsimonious model (the simplest model that still adequately fits the data). Specifically, a model that balances goodness-of-fit against model complexity to avoid the overfitting trap (Section 3.5.2).

To be able to achieve this goal, we begin by proposing a baseline model including all available environmental and temporal covariates. We follow the methods for automated variable selection (Section 6.4). Then we apply a stepwise regression procedure (step in R). This algorithm searches through the potential subsets of covariates, adjusting variables to find the most optimal model that minimises the Akaike Information Criterion (AIC). Finally, we use formal frequentist hypothesis test to confirm our selection.

Model Justification

To determine the most appropriate model, an automatic stepwise regression procedure, searching in both directions, was utilised (Section 6.4). This algorithm systematically searches through the covariates and finds the model that minimises the AIC. This, in turn, successfully balances the goodness-of-fit against model complexity, therefore ensuring we find the most parsimonious model.

The algorithm opted to drop atemp and workingday (as seen in the final model summary). This decision makes strong statistical sense because, for example, "feeling temperature" (atemp) and "actual temperature" (temp) provide extremely similar results. Therefore keeping both would increase the model complexity and introduce multicollinearity without adding meaningful explanatory power.

To be able to rigorously confirm the automated selection, we will run a formal hypothesis test to compare the nested "bestmodel" vs the "model" (the full model). We will be using these following hypothesis:

$H_0 : \beta_{\text{atemp}} = \beta_{\text{workingday}} = 0$(The new nested model is adequate. The dropped variables have no explanatory effect)

$$H_1 : \text{At least one of the coefficients is non zero (the full model is needed)}$$

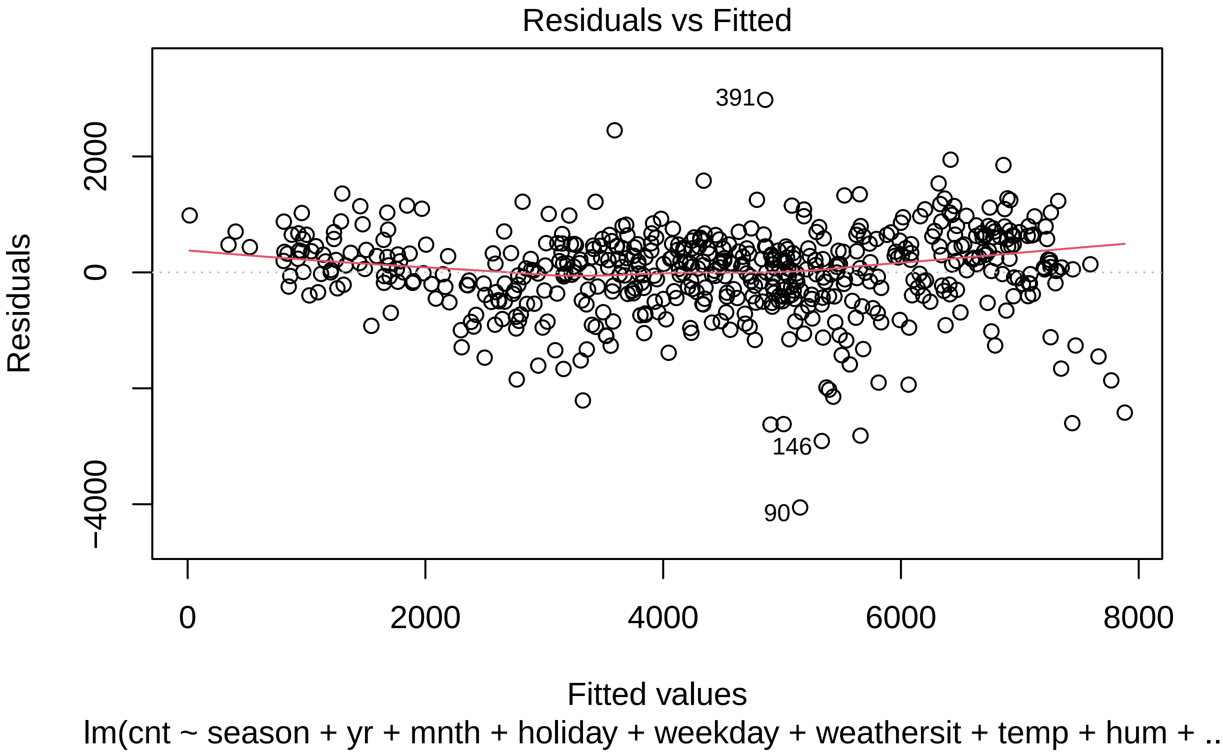
From the Analysis of Variance Table generated above we can see that the resulting F-test yielded a p-value of 0.2953. This value is significantly greater than the standard significance level of 0.05. Therefore, we do not have sufficient evidence to reject the null hypothesis (H_0). This, furthermore, provides formal statistical evidence that the variables that were chosen to be dropped (atemp and workingday) contribute no significant explanatory power, confirming that our parsimonious model is mathematically justified.

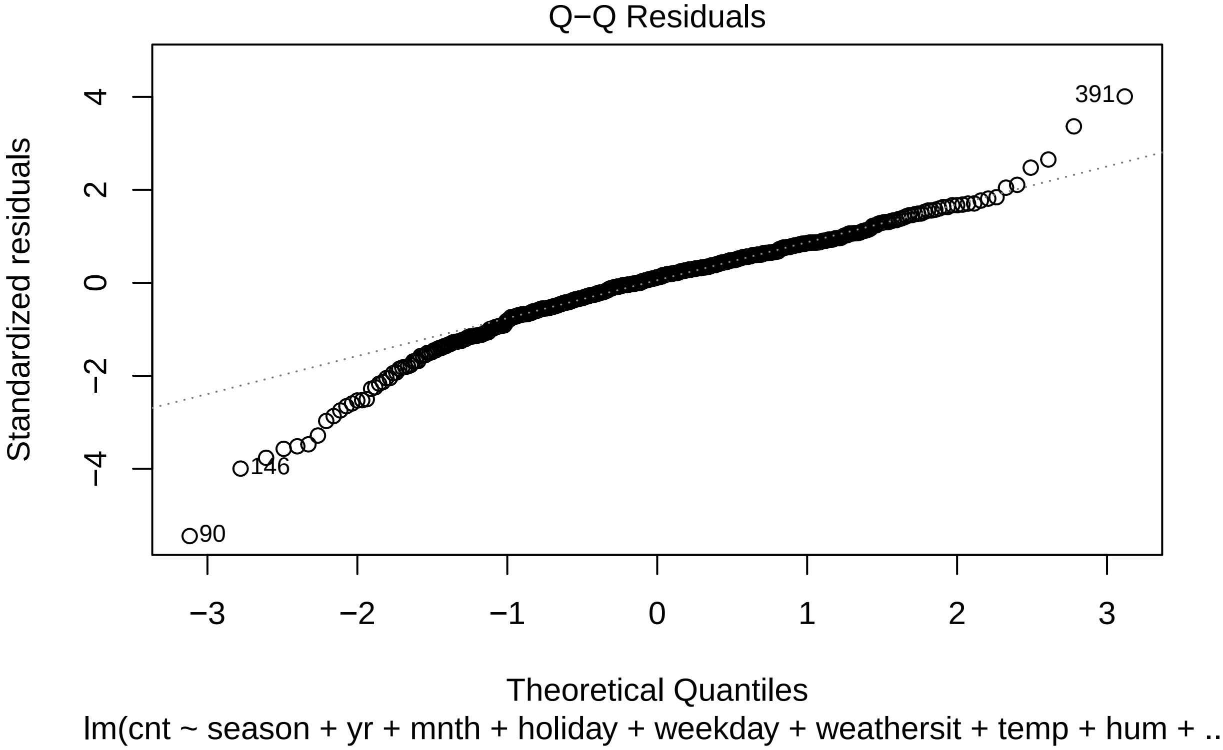
Finally, to evaluate the overall fit of the model, we examine the Adjusted R^2. In the case of this final selected model it is 0.841 as seen in the summary output. We will look at the Adjusted R^2 rather than the standard R^2 (Section 3.5.1). This is because, while the standard R^2 artificially inflated whenever any variable is added, the adjusted R^2 penalises the addition of unnecessary parameters. the adjusted R^2. therefore can be interpreted as approximately 84.1% of the variance in the daily rental bike counts is successfully explained by our final parsimonious model. This, therefore, capturing the true underlying trend while avoiding the overfitting trap (Section 3.5.2).

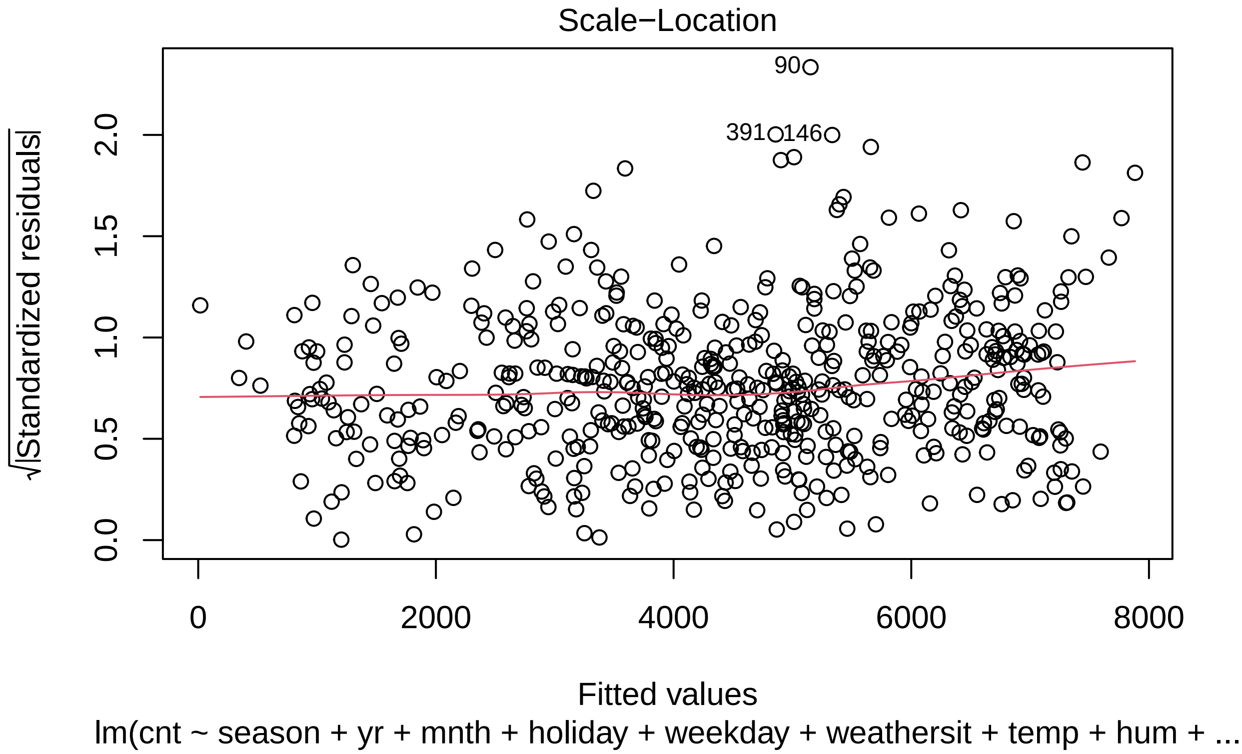
Model Checking and Discussion

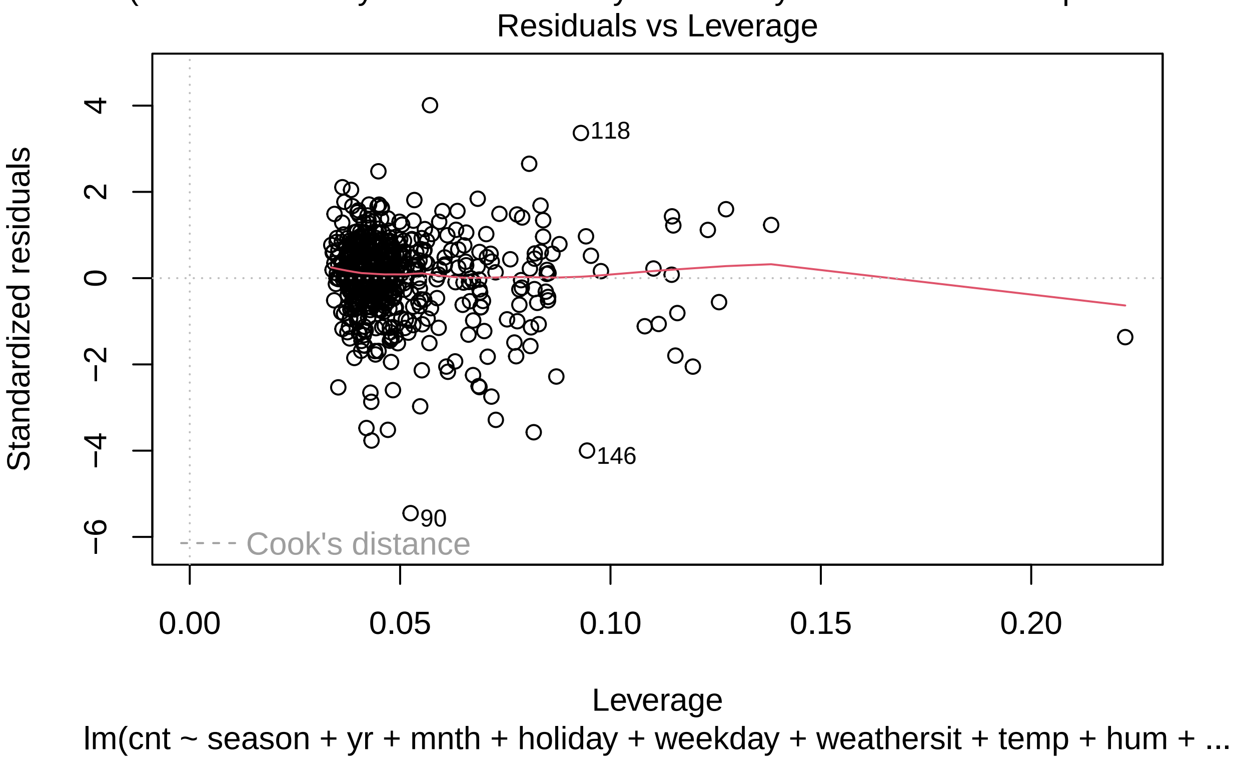
Model Diagnostics

To check that our chosen parsimonious multiple linear model is valid and reliable for prediction, we need to ensure that the core assumptions of normal linear models hold true (Chapter 6). We will plot the "best model" to check this. These assumptions include constant variance of the residuals and normality of the error terms (Section 6.2). Crucially, it is also important to check for any highly influential data point or outliers that could be disproportionately skewing the model fit (this is usually assessed via Cook's distance). The diagrams are presented below.


Residuals vs Fitted
Residuals
Fitted values
lm(cnt ~ season + yr + mnth + holiday + weekday + weathersit + temp + hum + ...
391
146
90
2000
0
−4000
0
2000
4000
6000
8000


Q−Q Residuals
Standardized residuals
Theoretical Quantiles
lm(cnt ~ season + yr + mnth + holiday + weekday + weathersit + temp + hum + ...
391
146
90
4
2
0
−2
−4
−3
−2
−1
0
1
2
3





Looking at the Residual vs Fitted plot and the Scale-Location, we can see that the points mostly present a random scatter mostly near the y=0 point. This suggests that the assumption of constant variance is generally satisfied (Section 6.2). Now looking at the Q-Q Residuals plot, we can note that the points are mostly falling on the dotted line with some of the points at each end deviating away from it. However, as most of the residuals fall along the dashed line, there is no strong evidence to suggest the residuals are not normally distributed (Section 6.2). Finally, the Residuals vs Leverage plot, we can see there are no significant outliers exceeding the standard Cook's distance threshold suggesting that there are no highly influential points materially distorting the regression coefficients (Section 6.3).

Parameter Interpretation

After checking the model to see if it is adequate, we can interpret the estimated coefficients to understand the real-world driving factors of the demand for rental bikes. We are looking to identify which parameters are the most important for the predictors. By examining the significance codes in the model summary, we can identify the strongest predictors as those with a p-value near zero ($p < 0.001$).

The most significant predictors, as we can see from the summary, are the normalised temperature (temp), humidity (hum) and windspeed. We can see that the coefficient for temp is strongly positive (4563.43). This is the highest coefficient out of the three, therefore it indicates that, while holding all other covariates constant, the warmer the weather the higher the demand for rented bike. This can also be further supported by the box plot from the EDA, we can see that the median of total rentals for bikes in the warmer months is higher. On the other hand, the windspeed has the highest negative coefficient (-2915.27) out of the three. This implies that windier days significantly decrease the demand for rental bikes.

There are also categorical factors that are of importance for predictors. We can see that the weather situation (weathersit) are highly significant. We can see that the "weathersit3" parameter (light Snow/Rain) has a very high negative coefficient (-1883.16). This indicates that the demand for rental bikes during those weather conditions is less. This again can be supported from the box plots in the EDA as the median for total rentals during winter and spring (where these weather conditions are common according to BBC weather) are lower than during the summer and fall. Additionally, the yr1 (Year 2012) parameter also has a strongly positive coefficient (2036.70). This demonstrates a large growth in the Capital Bikeshare system's popularity and usage between 2011 and 2012.

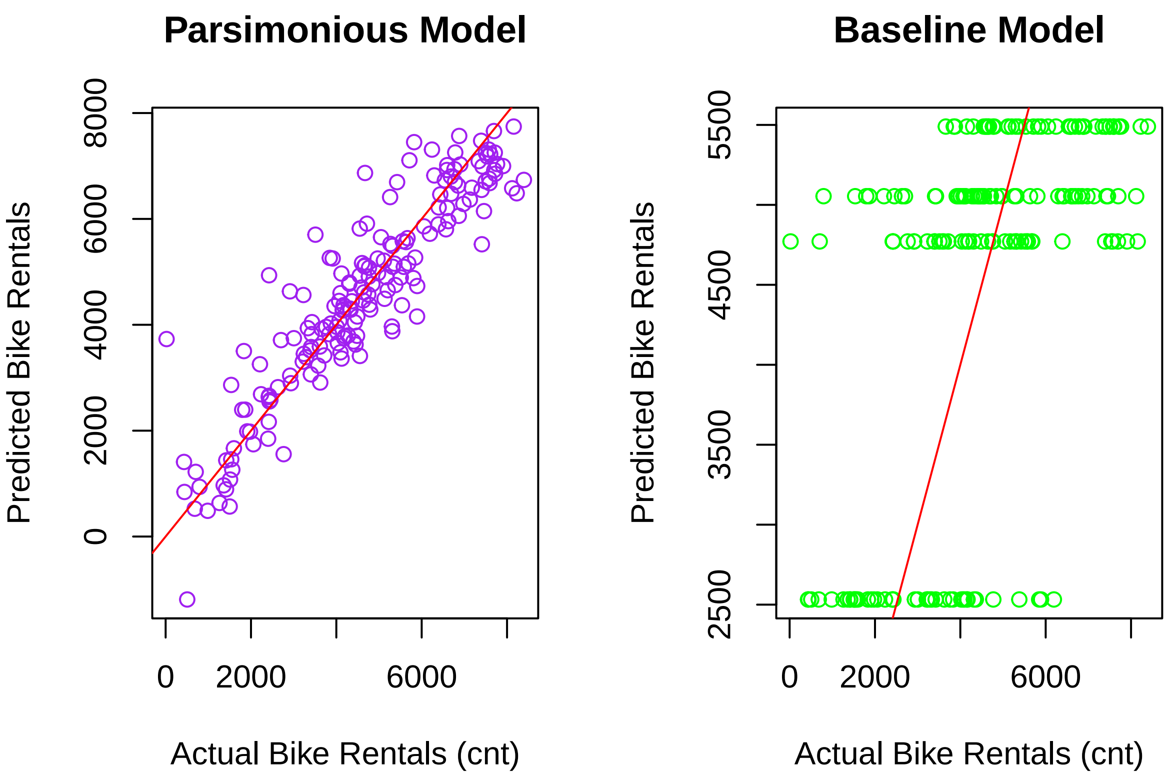
Part (b):

To test and evaluate the predictive performance of our selected multiple linear regression model on the new, unseen data, we will generate the predictions of the total count of rental bikes (cnt) using the new independent data set (BikeTest.csv). Then, we will compare the performance of our parsimonious model from part(a) against a simple baseline linear model of the form cnt ~ season.

To evaluate the quality of these predictions, we will calculate the Test Deviance (Residual Sum of Squares) for both of our models as our summary. Deviance measures the sum of squared predictor errors (Definition 3.3). Therefore, a lower Test Deviance will indicate a better fit to the unseen data. Plots will also be provided to visually assess the accuracy of both models.

```
## Test Deviance of Parsimonious Model:  115705948
```

```
## Test Deviance of Baseline Model:  494819527
```



Based on the calculations and plots above, our parsimonious model greatly outperforms the baseline model. Firstly, examining the Test Deviance scores generated, we can see that the parsimonious model achieved a Test Deviance of 115,705,948. This is substantially lower than the Test Deviance of the baseline model (494,819,527). This superiority is also clearly visible in the plots. Looking at the plot for the parsimonious model, we can see that the predicted values are very close to the red diagonal line, representing perfect prediction. In contrast, the plot for the baseline model fails to capture the true variance in daily bike rentals. Its predictions are strictly limited to four seasonal averages, resulting in a massive vertical spread of actual values that deviate greatly from the red line. In conclusion, we can confidently say that our parsimonious model successfully captures the true underlying trends and is highly effective for predicting daily bike rental demand.

#Appendix

```r
## Part(a):
library(ggplot2)
library(gridExtra)

train_data <- read.csv("BikeTrain.csv")
test_data <- read.csv("BikeTest.csv")

category_variables <- c("season", "yr", "mnth", "holiday", "weekday",
              "workingday", "weathersit")

train_data[category_variables] <- lapply(train_data[category_variables], as.factor)
test_data[category_variables] <- lapply(test_data[category_variables], as.factor)


plot1 <- ggplot(train_data, aes(x= temp,y = cnt)) +
  geom_point(a=0.5) +
  geom_smooth(method = "lm", col = "blue", se = FALSE) +
  theme_bw() +
  labs(title = "Bike Rentals vs Temperature",
       x = "Normalised Temperature",
       y = "Total Rentals")

plot2 <- ggplot(train_data, aes(x = season, y = cnt, fill = season)) +
  geom_boxplot() +
  theme_bw() +
  labs(title = "Bike Rentals by Season",
       x = "Season (1 = Spring, 2 = Summer, 3 = Fall, 4 = Winter) ",
       y = "Total Rentals") +
  theme(legend.position = "none")
grid.arrange(plot1, plot2, ncol = 2)



model <- lm(cnt ~ season + yr + mnth + holiday + weekday + workingday + weathersit +
              temp + atemp + hum + windspeed, data = train_data)

bestmodel <- step(model, direction = "both", trace = 0)

summary(bestmodel)

anova(bestmodel, model)




plot(bestmodel)




bestpred <- predict(bestmodel, newdata = test_data)
baselinemodel <- lm(cnt ~ season, data = train_data)
predbaseline <- predict(baselinemodel, newdata = test_data)
```

```r
devbest <- sum((test_data$cnt - bestpred)^2)
devbaseline <- sum((test_data$cnt - predbaseline)^2)

cat("Test Deviance of Parsimonious Model: ", devbest, "\n")
cat("Test Deviance of Baseline Model: ", devbaseline, "\n")
par(mfrow = c(1, 2))
plot(test_data$cnt, bestpred,
     main = "Parsimonious Model",
     xlab = "Actual Bike Rentals (cnt)",
     ylab = "Predicted Bike Rentals",
     col = "purple")
abline(0, 1, col = "red")

plot(test_data$cnt, predbaseline,
     main = "Baseline Model",
     xlab = "Actual Bike Rentals (cnt)",
     ylab = "Predicted Bike Rentals",
     col = "green")
abline(0, 1, col = "red")
```